

## Test Cases — SCWC Industry 5.0 Controller

Deterministic validation vectors exercised against Source\_Code/controller.c. All 10/10 test cases PASS.

| ID  | Description                        | Temp(C ) | Vib(mm/s) | Prox(cm ) | Power(W) | Expected State | Actual State | Result |
|-----|------------------------------------|----------|-----------|-----------|----------|----------------|--------------|--------|
| TC0 | Nominal operation                  | 42.0     | 1.2       | 150.0     | 320.0    | NORMAL         | NORMAL       | PASS   |
| TC1 | Temperature warning boundary       | 62.5     | 1.5       | 150.0     | 340.0    | WARNING        | WARNING      | PASS   |
| TC2 | Temperature critical               | 85.0     | 2.0       | 150.0     | 360.0    | CRITICAL       | CRITICAL     | PASS   |
| TC3 | Vibration warning boundary         | 45.0     | 4.5       | 150.0     | 330.0    | WARNING        | WARNING      | PASS   |
| TC4 | Vibration critical (bearing wear)  | 46.0     | 8.2       | 150.0     | 335.0    | CRITICAL       | CRITICAL     | PASS   |
| TC5 | Human proximity warning zone       | 40.0     | 1.1       | 55.0      | 310.0    | WARNING        | WARNING      | PASS   |
| TC6 | Human proximity DANGER (E-stop)    | 41.0     | 1.0       | 18.0      | 310.0    | SAFETY_STOP    | SAFETY_STOP  | PASS   |
| TC7 | Multi-fault + safety priority      | 90.0     | 7.5       | 12.0      | 355.0    | SAFETY_STOP    | SAFETY_STOP  | PASS   |
| TC8 | Idle -> low power / sustainability | 30.0     | 0.2       | 400.0     | 3.0      | LOW_POWER      | LOW_POWER    | PASS   |
| TC9 | Exact-threshold edge case          | 60.0     | 4.0       | 60.0      | 325.0    | WARNING        | WARNING      | PASS   |

### Notes:

- TC6 and TC7 confirm the safety interlock (proximity  $\leq 30$  cm) always takes priority over machine-health alarms, satisfying the human-centricity requirement of Industry 5.0.
- TC9 is an edge case placed exactly on the temperature/vibration warning boundary to verify the  $\geq$  comparison operators behave as specified.
- TC8 verifies the sustainability / low-power branch triggers only when the cell is both unloaded (power  $\leq 5$ W) and unoccupied (proximity  $>$  warning distance).